

Multiple Choice

1. For a certain complex number z , where $\arg(z)=\pi/5$, then $\arg(z^7)$ is

- A. $2\pi/5$
- B. $3\pi/5$
- C. $-7\pi/5$
- D. $-3\pi/5$

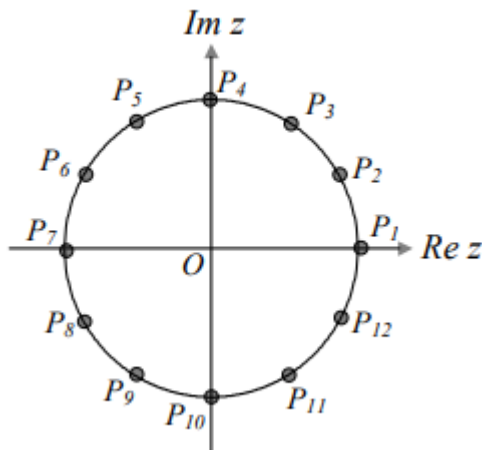
2. Consider the complex numbers $2z$, iz , and $2z - iz$, where $z \in \mathbb{C}$, $z \neq 0$.

These three complex numbers are plotted in the Argand plane and together with the origin O they form the vertices of a quadrilateral.

The area of this quadrilateral is

- A. $\frac{1}{|2z|}$
- B. $|2z|$
- C. $|z^2|$
- D. $2|z|^2$

3. The points P_1 to P_{12} are twelve equally spaced points around the circumference of a circle.



Point P_3 represents the complex number $z = a + ib$. The complex number $i^{11}\bar{z}$ is represented by point

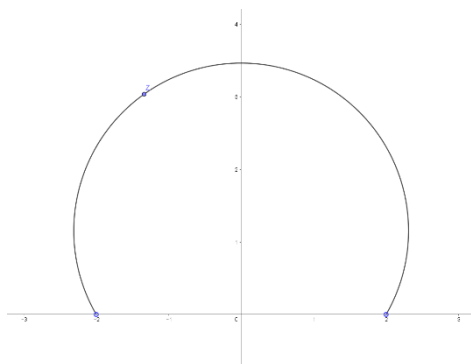
- A. P_5
- B. P_8
- C. P_9
- D. P_{11}

4. It is known that ω is a 4th root of $\frac{\sqrt{7}}{\sqrt{10}} + \frac{\sqrt{3}}{\sqrt{10}}i$. Which of the following is also a root?

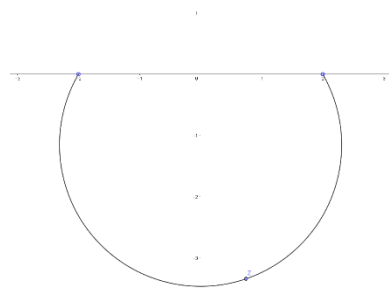
- A. ω^2
- B. $-\omega$
- C. $\bar{\omega}$
- D. $-\bar{\omega}$

5. Which of the following represents the locus of z such that $\text{Arg}\left(\frac{z+2}{z-2}\right) = \frac{2\pi}{3}$?

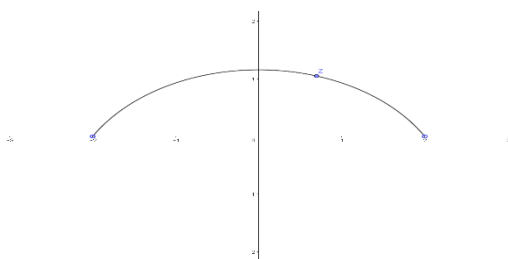
A



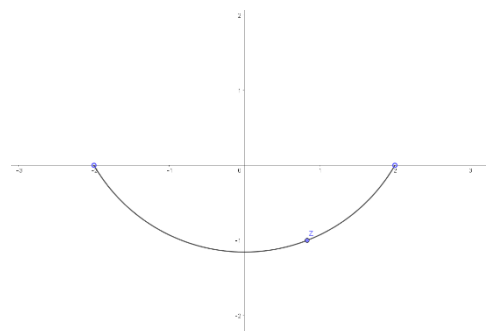
B



C



D



Question 6 (9 marks)

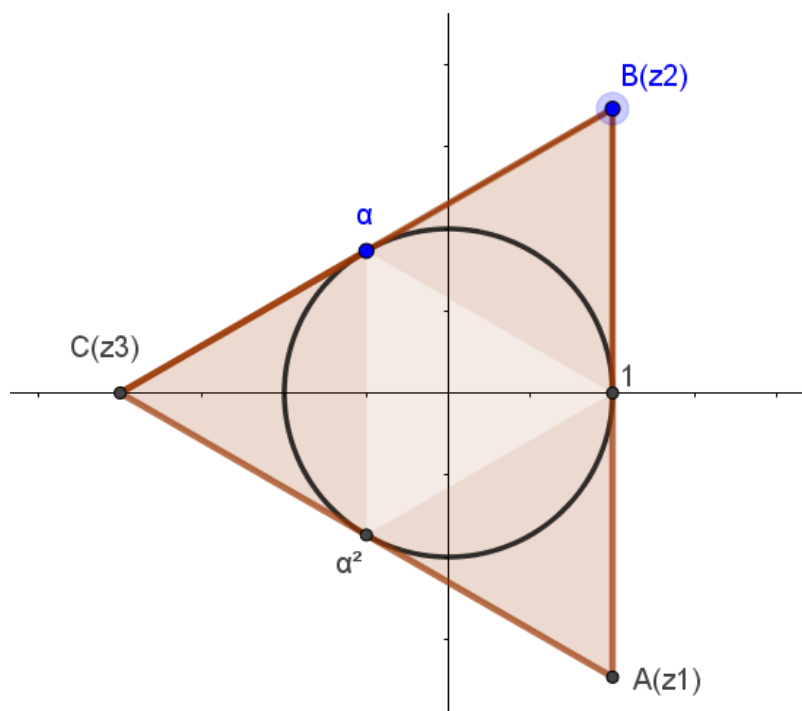
- A. Given $z = 2 + 3i$ and $w = 3 + i$, find $\overline{z + iw}$ in the form $a + ib$ 2m
- B. Given that $3 + i$ is a root to the equation $z^4 - 4z^3 + 3z^2 - 10z + 50 = 0$, factorise $z^4 - 4z^3 + 3z^2 - 10z + 50$ into:
- I. Factors with real coefficients 2m
 - II. Linear Factors 2m
- C. On an Argand Diagram, sketch the locus of z such that $\text{Arg}(z + 1 + i) = \text{Arg}(z - 2i)$ 3m

Question 7 (Start a New Page) (9 marks)

- A. Solve the equation $z^2 = |z|^2 - 4$ 3m
- A. i) Show that $3 - i\sqrt{3}$ is $2\sqrt{3} \left(\cos\left(-\frac{\pi}{6}\right) + i \sin\left(-\frac{\pi}{6}\right) \right)$. 1m
- ii) Find $(3 - i\sqrt{3})^6$ and express your solution in the form $x + iy$ where $x, y \in \mathbb{R}$. 2m
- iii) Find the integer values of n for which $(3 - i\sqrt{3})^n$ is real. 1m
- iv) Find the integer values of n for which $(3 - i\sqrt{3})^n = ai$, where a is a real number. 1m
- B Express $e^{\frac{i\pi}{3}}$ in Cartesian form. 1m

Question 8 (Start a New Page) (8 marks)

Given that $1, \alpha$ and α^2 are three complex numbers on the Argand diagram, where α is a non-real cube root of unity.



- i. Show that $1 + \alpha + \alpha^2 = 0$. 1m
- ii ABC is an equilateral triangle, z_1, z_2 and z_3 are the complex numbers corresponding to A, B and C respectively. Write down z_1, z_2 and z_3 in terms of α . 3m
- iii. Show that $z_2 + \alpha z_3 + \alpha^2 z_1 = 0$. 1m
- iv Let $\omega_1 = \frac{1}{3}(1 + \alpha + z_2)$, $\omega_2 = \frac{1}{3}(\alpha + \alpha^2 + z_3)$ and $\omega_3 = \frac{1}{3}(1 + \alpha^2 + z_1)$ 3m
be three complex numbers, prove that ω_1, ω_2 and ω_3 are the cube roots of -1.

Question 9 (Start a New Page) (10 marks)

Let $v = 2 + 2\sqrt{3}i$ and $w = 2\sqrt{2} + 2\sqrt{2}i$.

- i) If $z = \frac{v}{w}$ give z in modulus-argument form. 3m
- ii) On an Argand diagram, plot v, w and z . 2m
- iii) Using your results from part (iii), show that $\tan\left(\frac{\pi}{12}\right) = 2 - \sqrt{3}$. 2m
- iv) Given that $z = \frac{v}{w}$ is a solution of the equation

$$z^3 + \frac{1}{2}(\sqrt{2} - 3\sqrt{6})z^2 + 3z - \sqrt{6} + \sqrt{2} = 0,$$
find the other two solutions of the equation, expressing your answers in Cartesian form. 3m

Question 10 (Start a New Page) (11 marks)

A Given that $z = 2^{\frac{1}{3}}\left(\cos\frac{\pi}{4} - i\sin\frac{\pi}{4}\right)$,

- i) Show that $z^3 + \frac{1}{z^3} = \frac{-5}{2\sqrt{2}} - \frac{3}{2\sqrt{2}}i$ 3m
- ii) Find the real numbers a and b if $z^3 + \frac{1}{z^3} = e^a \cdot e^{ib}$ 3m

B Given $|z_1| = |z_2| = 2$, and $\arg(z_1) = \frac{\pi}{5}$ and $\arg(z_2) = \frac{3\pi}{5}$,

Show that $(z_1 - z_2) = 4\sin\left(\frac{\pi}{5}\right)\text{cis}\left(\frac{-\pi}{10}\right)$ 5m

END of PAPER

MATHEMATICS Extension 2: Question. MC...

Suggested Solutions

Marks

Marker's Comments

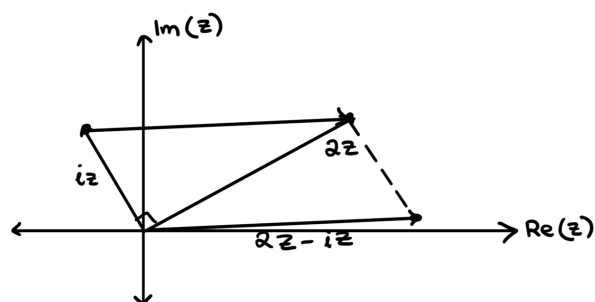
1. D

$$\arg(z) = \frac{\pi}{5}$$

$$\arg(z^7) = \frac{7\pi}{5} \quad (\text{De Moivre's theorem})$$

$$\begin{aligned} \text{Arg}(z^7) &= \frac{7\pi}{5} - 2\pi \\ &= -\frac{3\pi}{5} \end{aligned}$$

2. D



$$\begin{aligned} \text{Area} &= 2 \times \frac{1}{2} \times |iz| \times |2z| \\ &= |z| \times 2 \times |z| \\ &= 2|z|^2 \end{aligned}$$

3. B

$$z = P_3$$

$$\bar{z} = P_{11}$$

$$i^4 \bar{z} = P_{11}$$

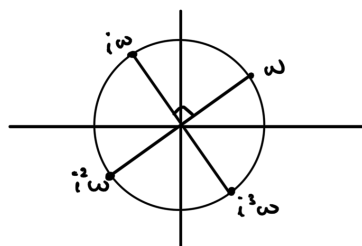
$$i^8 \bar{z} = P_{11}$$

$$i^9 \bar{z} = P_2$$

$$i^{10} \bar{z} = P_5$$

$$i^{11} \bar{z} = P_8$$

4. B



4th roots are equally spaced, $\frac{\pi}{2}$ apart.

$$\therefore i^2 \omega = -\omega.$$

5. D

MATHEMATICS Extension 2: Question.....7

Suggested Solutions

Marks

Marker's Comments

i) 1, α and α^2 are the roots of the equation $z^3 - 1 = 0$

①

$$\therefore 1 + \alpha + \alpha^2 = \frac{-b}{a} \quad (\text{Sum of roots}) \leftarrow \text{necessary!}$$

$$= \frac{-0}{1}$$

$$= 0$$

①

Alternatively, some students solved for α and α^2 , and proved it by actually doing the sum.

or

Since α is a root to $z^3 - 1 = 0$

$$\text{then } \alpha^3 - 1 = 0$$

$$(\alpha - 1)(\alpha^2 + \alpha + 1) = 0$$

Since α is non-real, $\alpha \neq 1$ (necessary!)

$$\therefore \alpha^2 + \alpha + 1 = 0$$

①

①

ii) $z_1 = \alpha^2 - \alpha + 1$

①

$$z_2 = -\alpha^2 + \alpha + 1$$

①

$$z_3 = \alpha^2 + \alpha - 1$$

①

Note: $1 + \sqrt{3}i$ is not considered to be "in terms of α ".

iii) $(-\alpha^2 + \alpha + 1) + \alpha(\alpha^2 + \alpha - 1) + \alpha^2(\alpha^2 - \alpha + 1)$

$$= -\alpha^2 + \alpha + 1 + \alpha^3 + \alpha^2 - \alpha + \alpha^4 - \alpha^3 + \alpha^2$$

$$= \alpha^4 + \alpha^2 + 1$$

$$= \alpha^2 + \alpha + 1 \quad (\text{since } \alpha^3 = 1)$$

$$= 0 \quad (\text{proven from part i})$$

①

MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks

Marker's Comments

$$\begin{aligned} \text{iv) } W_1 &= \frac{1}{3}(1 + \omega + (-\omega^2 + \omega + 1)) \\ &= \frac{1}{3}(2 + 2\omega - \omega^2) \\ &= \frac{1}{3}[2(1 + \omega + \omega^2) - 3\omega^2] \\ &= \frac{1}{3}(-3\omega^2) \\ &= -\omega^2 \end{aligned}$$

$$\begin{aligned} W_1^3 &= -\omega^6 \\ &= -(1)(1) \\ &= -1 \end{aligned}$$

$$\begin{aligned} W_2 &= \frac{1}{3}(\omega + \omega^2 + (\omega^2 + \omega - 1)) \\ &= \frac{1}{3}(2\omega^2 + 2\omega - 1) \\ &= \frac{1}{3}[2(\omega^2 + \omega + 1) - 3] \\ &= \frac{1}{3}(-3) \\ &= -1 \end{aligned}$$

$$\begin{aligned} W_2^3 &= (-1)^3 \\ &= -1 \end{aligned}$$

$$\begin{aligned} W_3 &= \frac{1}{3}(1 + \omega^2 + \omega^2 - \omega + 1) \\ &= \frac{1}{3}(2\omega^2 - \omega + 2) \\ &= \frac{1}{3}[2(\omega^2 + \omega + 1) - 3\omega] \\ &= \frac{1}{3}(-3\omega) \\ &= -\omega \end{aligned}$$

$$\begin{aligned} W_3^3 &= (-\omega)^3 \\ &= -\omega^3 \\ &= -1 \end{aligned}$$

$\therefore W_1, W_2$ and W_3 are the cube roots of -1 .
(Since W_1, W_2 and W_3 are distinct points)

① for solving one of W_1, W_2 or W_3 in simplified form

① for solving the other 2

① Proving that the cube of all 3 is -1 .

MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks

Marker's Comments

iv) For those who wanted to prove that w_1, w_2 and w_3 are roots to the polynomial $z^3 + 1 = 0$

① Proving $w_1 + w_2 + w_3 = 0$
AND
 $w_1 w_2 w_3 = -1$

① Attempt to prove that $w_1 w_2 + w_1 w_3 + w_2 w_3 = 0$ and making significant progress.

① Doing the first 2 steps successfully and finishing the rest of the proof with sound argument.

MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks

Marker's Comments

Accepted answers: ii)

z_1	z_2	z_3
$d^2 - d + 1$	$-d^2 + d + 1$	$d^2 + d - 1$
$d^2(1 + \sqrt{3}i)$		$d(1 + \sqrt{3}i)$
	$d(1 - \sqrt{3}i)$	$d^2(1 - \sqrt{3}i)$
$-2d$	$-2d^2$	
$2d^{\frac{1}{2}}$	$2d^{\frac{1}{2}}$	$2d^{\frac{3}{2}}$
$2d^2 e^{\frac{i\pi}{3}}$		$2d e^{\frac{i\pi}{3}}$
	$2d e^{-\frac{i\pi}{3}}$	$2d^2 e^{-\frac{i\pi}{3}}$

Question 9,

(a) (i) show that $3 - i\sqrt{3}$ is $2\sqrt{3} \left(\cos\left(-\frac{\pi}{6}\right) + i \sin\left(-\frac{\pi}{6}\right) \right)$

$$\begin{aligned} 3 - i\sqrt{3} &= 2\sqrt{3} \left(\frac{\sqrt{3}}{2} + i \left(\frac{-1}{2} \right) \right) \\ &= 2\sqrt{3} \left(\cos\left(-\frac{\pi}{6}\right) + i \sin\left(-\frac{\pi}{6}\right) \right) \end{aligned}$$

(ii) Find the integer values of n for which $(3 - i\sqrt{3})^n = ai$, where a is a real number.

$$(3 - i\sqrt{3})^n = ai$$

$$\therefore \left[2\sqrt{3} \left(\cos\left(-\frac{\pi}{6}\right) + i \sin\left(-\frac{\pi}{6}\right) \right) \right]^n = ai$$

$$= (2\sqrt{3})^n \left[\cos\left(-\frac{n\pi}{6}\right) + i \sin\left(-\frac{n\pi}{6}\right) \right] = ai \quad \text{by de Moivre's Theorem}$$

$$\therefore \cos \frac{n\pi}{6} = 0 \quad \therefore \frac{n\pi}{6} = \frac{(2k+1)\pi}{2}, \quad k \text{ is an integer}$$

$$\begin{aligned} n &= 3(2k+1) \\ &= 6k+3, \quad k \in \mathbb{Z} \end{aligned}$$

$$? \quad \underline{\underline{-6k-3}}$$

(b) Given that $z = 2^{\frac{1}{3}} \left(\cos \frac{\pi}{4} - i \sin \frac{\pi}{4} \right)$

(i) Show that $z^3 + \frac{1}{z^3} = \frac{-5}{2\sqrt{2}} - \frac{3}{2\sqrt{2}}i$

$$z^3 = 2 \left(\cos \frac{3\pi}{4} - i \sin \frac{3\pi}{4} \right) \quad \text{by de Moivre's theorem}$$

$$\frac{1}{z^3} = \frac{1}{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right) \quad \text{by de Moivre's theorem}$$

$$z^3 + \frac{1}{z^3} = \frac{5}{2} \cos \frac{3\pi}{4} - \frac{3}{2} \sin \frac{3\pi}{4} i$$

$$= \frac{5}{2} \left(\frac{-\sqrt{2}}{2} \right) - \frac{3}{2} \left(\frac{\sqrt{2}}{2} \right) i$$

$$= \frac{-5\sqrt{2}}{4} - \frac{3\sqrt{2}}{4} i$$

$$= \frac{-5}{2\sqrt{2}} - \frac{3}{2\sqrt{2}} i$$

Alt

$$\left. \begin{aligned} 2^{-\frac{1}{3}} z &= \cos \frac{\pi}{4} - i \sin \frac{\pi}{4} \\ \frac{2^{\frac{1}{3}}}{z} &= \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \end{aligned} \right\} \therefore \begin{aligned} \frac{z^3}{2} &= \cos \frac{3\pi}{4} - i \sin \frac{3\pi}{4} \\ \frac{2}{z^3} &= \cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \end{aligned} \quad \text{By de Moivre's theorem}$$
$$z^3 = 2 \left(\cos \frac{3\pi}{4} - i \sin \frac{3\pi}{4} \right) \quad \text{and} \quad \frac{1}{z^3} = \frac{1}{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$$

(ii)

$$z^3 + \frac{1}{z^3} = e^a \cdot e^{ib} = e^a (\cos b + i \sin b)$$

$$e^a \cos b = \frac{-5}{2\sqrt{2}} \quad \text{and} \quad e^a \sin b = \frac{-3}{2\sqrt{2}}$$

$$e^{2a} = \frac{25+9}{8} = \frac{34}{8} = \frac{17}{4} \therefore e^a = \frac{\sqrt{17}}{2} \therefore a = \ln\left(\frac{\sqrt{17}}{2}\right)$$

$$a = \frac{1}{2} \ln(17) - \ln 2$$

$$\cos b = \frac{-5}{\sqrt{34}}, \quad \sin b = \frac{-3}{\sqrt{34}}$$

$$b = \tan^{-1}\left(\frac{3}{5}\right) - \pi$$

$$? \tan^{-1}\left(\frac{3}{5}\right)$$

$$? \pi + \tan^{-1}\left(\frac{3}{5}\right)$$

$$? -\pi + \tan^{-1}\left(\frac{3}{5}\right) + 2k\pi, k \in \mathbb{Z}$$

(a)

(i) 1 for correct

0 for wrong

(ii) 1 for correct

0 for wrong.

$6k+3, k \in \mathbb{Z}$
or $-6k-3, k \in \mathbb{Z}$

(b)

(i) 1 for z^3

1 for $\frac{1}{z^3}$

1 for final correct answer

(ii)

1 for equating real and imaginary parts

1 for correct a

1 for correct b

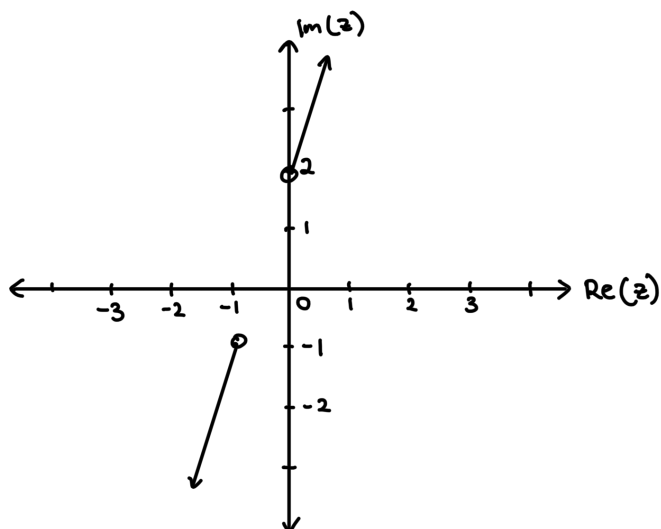
MATHEMATICS Extension 2: Question 10...

Suggested Solutions

Marks

Marker's Comments

a) $\text{Arg}(z + 1 + i) = \text{Arg}(z - 2i)$



1 correct points
1 open circles
1 correct lines

b) $|z_1| = |z_2| = 2$, $\arg(z_1) = \frac{\pi}{5}$ and $\arg(z_2) = \frac{3\pi}{5}$

$$z_1 = 2 \left(\cos \frac{\pi}{5} + i \sin \frac{\pi}{5} \right)$$

$$z_2 = 2 \left(\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5} \right)$$

$$\begin{aligned} z_1 - z_2 &= 2 \left(\cos \frac{\pi}{5} - \cos \frac{3\pi}{5} + i \left(\sin \frac{\pi}{5} - \sin \frac{3\pi}{5} \right) \right) \\ &= 2 \left[\cos \left(\frac{2\pi}{5} - \frac{\pi}{5} \right) - \cos \left(\frac{2\pi}{5} + \frac{\pi}{5} \right) \right. \\ &\quad \left. + i \left(\sin \left(\frac{2\pi}{5} + \left(-\frac{\pi}{5} \right) \right) - \sin \left(\frac{2\pi}{5} - \left(-\frac{\pi}{5} \right) \right) \right) \right] \\ &= 2 \left[2 \sin \left(\frac{2\pi}{5} \right) \sin \left(\frac{\pi}{5} \right) + 2i \cos \left(\frac{2\pi}{5} \right) \sin \left(-\frac{\pi}{5} \right) \right] \\ &= 4 \left(\sin \frac{2\pi}{5} \sin \frac{\pi}{5} - i \cos \frac{2\pi}{5} \sin \frac{\pi}{5} \right) \end{aligned}$$

as $\sin(-x) = -\sin x$

$$= 4 \sin \frac{\pi}{5} \left(\sin \frac{2\pi}{5} - i \cos \frac{2\pi}{5} \right)$$

$$= 4 \sin \frac{\pi}{5} \left(\cos \left(\frac{\pi}{2} - \frac{2\pi}{5} \right) - i \sin \left(\frac{\pi}{2} - \frac{2\pi}{5} \right) \right)$$

$$= 4 \sin \frac{\pi}{5} \left(\cos \frac{\pi}{10} - i \sin \frac{\pi}{10} \right)$$

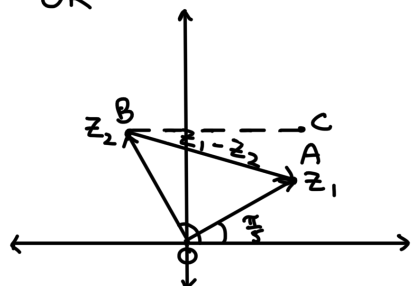
$$= 4 \sin \frac{\pi}{5} \left(\cos \left(-\frac{\pi}{10} \right) - i \sin \left(-\frac{\pi}{10} \right) \right)$$

using product to
sum on the
formula sheet.

must also show that
 $\sin x$ is an odd function

show working for
complementary angles

OR



$$\angle AOB = \frac{3\pi}{5} - \frac{\pi}{5}$$

$$= \frac{2\pi}{5}$$

$$OB = OA = 2$$

$\angle OBA = \angle OAB$ (equal angles
are opposite equal sides
in triangle)

correct diagram

MATHEMATICS Extension 2: Question...!O...

Suggested Solutions	Marks	Marker's Comments
$\angle OBA = \frac{\pi - \frac{2\pi}{5}}{2} \text{ (angle sum of a triangle)}$ $= \frac{3\pi}{10}$ $\angle CBO = \pi - \frac{3\pi}{5} \text{ (co-interior angles, parallel lines)}$ $= \frac{2\pi}{5}$ $\angle CBA = \frac{2\pi}{5} - \frac{3\pi}{10}$ $= \frac{\pi}{10}$ $\therefore \arg(z_1 - z_2) = -\frac{\pi}{10}$	1	for working out angle
$ z_1 - z_2 = \sqrt{ z_1 ^2 + z_2 ^2 - 2 z_1 z_2 \cos \angle AOB}$ $= \sqrt{4 + 4 - 8\cos \frac{2\pi}{5}}$ $= \sqrt{8 - 8\cos \frac{2\pi}{5}}$ $= \sqrt{8 - 8(1 - 2\sin^2 \frac{\pi}{5})}$ $= \sqrt{16\sin^2 \frac{\pi}{5}}$ $= 4\sin \frac{\pi}{5}$ $\therefore z_1 - z_2 = 4\sin \frac{\pi}{5} \left(\cos\left(-\frac{\pi}{10}\right) + i\sin\left(-\frac{\pi}{10}\right) \right)$	1	recognising arg is negative
		correct modulus plus conclusion.

Question 6

Sunday, 12 December 2021

10:11 pm

a) $z = 2 + 3i$, $w = 3 + i$

$$z + iw = 2 + 3i + i(3 + i)$$

$$= 2 + 3i + 3i - 1$$

$$= 1 + 6i$$

$$\overline{z + iw} = 1 - 6i$$

Im for $z + iw$

Im for $\overline{z + iw}$

Generally well done

b) $e^{i\pi/3} = \cos \pi/3 + i \sin \pi/3$

$$= \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

Im for Cartesian form
Om for mod-arg form

Generally well done.

Students must read question carefully and give answers in specified form.

c) $z = x + iy$

$$z^2 = |z|^2 - 4$$

$$(x + iy)^2 = (\sqrt{x^2 + y^2})^2 - 4$$

$$x^2 + 2xiy - y^2 = x^2 + y^2 - 4$$

Im equation in terms of x and y in expanded form

$$x^2 - y^2 = x^2 + y^2 - 4$$

$$-2y^2 = -4$$

$$y^2 = 2$$

$$y = \pm\sqrt{2}$$

Im for y

$$\therefore z = \pm\sqrt{2}i$$

Im for z

Answers must be given as z values

d) Since coefficients are real $3 - i$ is also a root by the conjugate root theorem

$$(z - (3 + i))(z - (3 - i))(z^2 + az + b) = z^4 - 4z^3 + 3z^2 - 10z + 50$$

$$(z^2 - 6z + 10)(z^2 + az + b) = z^4 - 4z^3 + 3z^2 - 10z + 50$$

$$-6 + a = -4$$

$$\therefore a = 2$$

$$10b = 50$$

$$b = 5$$

Students should practise factorisation by inspection.

$$\therefore z^4 - 4z^3 + 3z^2 - 10z + 50 = (z^2 - 6z + 10)(z^2 + 2z + 5)$$

Im for stating $3 - i$ is also a root

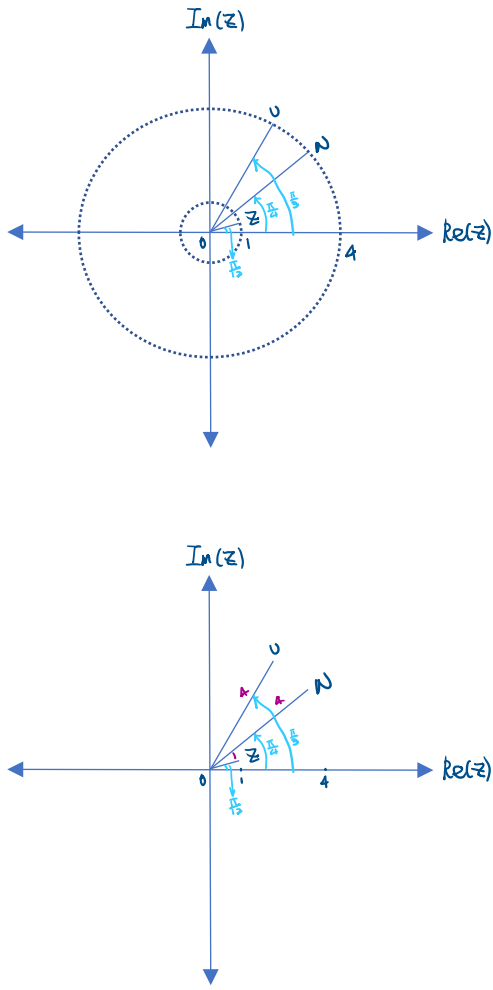
Im for $(z^2 - 6z + 10)$ as a factor

Im for $(z^2 + 2z + 5)$ as a factor And for not finding linear factors

2021 Year 12 Term 1 Mathematics Extension 2 Question 8

Solution	Marks	Marker's Comments
i) $z = \frac{2+2\sqrt{3}i}{2\sqrt{2}+2\sqrt{2}i} = \frac{1+\sqrt{3}i}{\sqrt{2}+\sqrt{2}i}$ $= \frac{(1+\sqrt{3}i)(1-i)}{\sqrt{2}(1+i)(1-i)}$ $= \frac{1+\sqrt{3}i-i+\sqrt{3}}{\sqrt{2}(1+1)}$ $= \frac{1+\sqrt{3}}{2\sqrt{2}} + \frac{\sqrt{3}-1}{2\sqrt{2}}i$	1 1	Multiplying the conjugate of the denominator "a+bi" form (some students write "$\frac{1+\sqrt{3}+(\sqrt{3}-1)i}{2\sqrt{2}}$" (x))
ii) $v = \sqrt{2^2 + (2\sqrt{3})^2} = 4$ $\text{Arg}(v) = \tan^{-1}\left(\frac{2\sqrt{3}}{2}\right) = \frac{\pi}{3}$ (fir quadrant) $v = 4\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$ $w = \sqrt{(2\sqrt{2})^2 + (2\sqrt{2})^2} = 4$ $\text{Arg}(w) = \tan^{-1}\left(\frac{2\sqrt{2}}{2\sqrt{2}}\right) = \frac{\pi}{4}$ first quad $w = 4\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$ $z = \frac{v}{w} = \frac{4\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)}{4\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)}$ $= \cos \frac{\pi}{12} + i \sin \frac{\pi}{12}$	1 1	Final answers need to be in the expanded form "$r(\cos \theta + i \sin \theta)$" correct v and w correct z (some students use modulus argument form to find the answer, which is really long)

2021 Year 12 Term 1 Mathematics Extension 2 Question 8

Solution	Marks	Marker's Comments
iii) 	1 1 1 1	* The complex numbers can be represented by Ⓐ a dot Ⓑ an interval Ⓒ an arrow * if one complex number is wrong, deduct one mark. arguments modulus (circle) arguments modulus (labeled) - u and w same length - z is shorter

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iii) continue	1 1	coordinate pairs v and w same length z is shorter
	1 1	coordinates labelled on the axis v and w same length z is shorter Common problems: ① label i — coordinate — y-axis ② no Im, Re ③ v and w different length

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iv) z is a solution for real coefficient equation $\Rightarrow \bar{z} = \frac{\sqrt{3}+1}{2\sqrt{2}} + \frac{1-\sqrt{3}}{2\sqrt{2}}i$ is also a solution let the third solution be β. Method 1:- product of roots = $z \cdot \bar{z} \cdot \beta = -\frac{d}{a} = -\frac{-(-\sqrt{6}+\sqrt{2})}{1} = \sqrt{6}-\sqrt{2}$ let $r \operatorname{cis} \theta = r(\cos \theta + i \sin \theta)$ $\operatorname{cis}\left(\frac{\pi}{12}\right) \times \operatorname{cis}\left(-\frac{\pi}{12}\right) \times \beta = \sqrt{6}-\sqrt{2}$ $\operatorname{cis}(0) \times \beta = \sqrt{6}-\sqrt{2}$ $1 \times \beta = \sqrt{6}-\sqrt{2}$ $\beta = \sqrt{6}-\sqrt{2}$ Method 2: Sum of roots = $z + \bar{z} + \beta = -\frac{b}{a}$ $2 \operatorname{Re}(z) + \beta = -\frac{1}{2}(\sqrt{2}-3\sqrt{6})$ $2 \times \frac{\sqrt{6}+\sqrt{2}}{4} + \beta = -\frac{1}{2}(\sqrt{2}-3\sqrt{6})$ $\times 2 \quad \left\{ \begin{array}{l} \sqrt{6}+\sqrt{2}+2\beta = -\sqrt{2}+3\sqrt{6} \\ 2\beta = -2\sqrt{2}+2\sqrt{6} \\ \beta = \sqrt{6}-\sqrt{2} \end{array} \right. \times 2$ Method 3: Change variable z to x $(x-z)(x-\bar{z})$ $= x^2 - 2 \operatorname{Re}(z)x + z\bar{z}$ $= x^2 - \left(\frac{\sqrt{2}+\sqrt{6}}{2}\right)x + \left(\frac{\sqrt{2}+\sqrt{6}}{4}\right)^2 + \left(\frac{\sqrt{6}-\sqrt{2}}{4}\right)^2$ $= x^2 - \left(\frac{\sqrt{2}+\sqrt{6}}{2}\right)x + \frac{2+6+2\sqrt{2}+2+6-2\sqrt{2}}{16}$ $= x^2 - \left(\frac{\sqrt{2}+\sqrt{6}}{2}\right)x + 1$ $(x^2 - \frac{\sqrt{2}+\sqrt{6}}{2}x + 1)(x-\beta) = x^3 + \frac{1}{2}(\sqrt{2}-3\sqrt{6})x^2 + 3x - \sqrt{6}+\sqrt{2}$ By equating the constant, $1 \times (-\beta) = -\sqrt{6}+\sqrt{2}$ $\beta = \sqrt{6}-\sqrt{2}$	1 1 1 1 1 1	The question asks for answers in cartesian form Some students forget the "-" of $-\frac{d}{a}$ Some students forget "-" of $-\frac{b}{a}$ A lot of calculation mistake here !!! Please don't rush !!! A lot of calculation mistakes.

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iv) continue Not recommended method let the other two solutions be α and β $\alpha + \beta + z = -\frac{1}{2}(\sqrt{5} - 3\sqrt{6})$ $\alpha\beta z = -(-\sqrt{6} + \sqrt{2})$ and then solve simultaneous equations to find out α and β, which involves too much calculation.	1 1 1	Find out α Find out β